

Evaluating the Impact of FOMC Communications on Asset Prices

FRE-GY 7871 A - NLP and the Investment Process

Forecast date: September 13, 2026

Executive summary

The sample contains 297 documents: FOMC statements and minutes, press-conference transcripts, and Chair speeches. Warsh-era communication is mostly positive on both tone measures, but the current Warsh sample remains small (nine documents). In the full sample, the word-list score predicts a positive growth-minus-value move (+0.083 percentage points, $p = 0.047$), while FinBERT predicts the opposite (-1.355 percentage points, $p = 0.058$). Neither method explains DXY, 10s2s, or the 1-year yield at conventional significance levels. The key result is therefore disagreement, not a robust directional signal.

signal.

My base case is a September hold and a 65% chance of a more hawkish statement. I retain a small long-IWF/short-IWN position because the lexicon result is statistically significant, but the trade is low conviction because FinBERT disagrees and 154 document timestamps need verification.

Data and methods

The sample begins in February 2018. Powell is the baseline Chair through May 21, 2026; the Warsh period begins May 22, 2026. Documents come from Federal Reserve calendars, historical materials, press-conference transcripts, and speech archives. The one-day outcomes are DXY, 10s2s, the 1-year Treasury yield, and IWF-minus-IWN. Regressions include the DGS3MO change, separating text-tone associations from contemporaneous short-rate news.

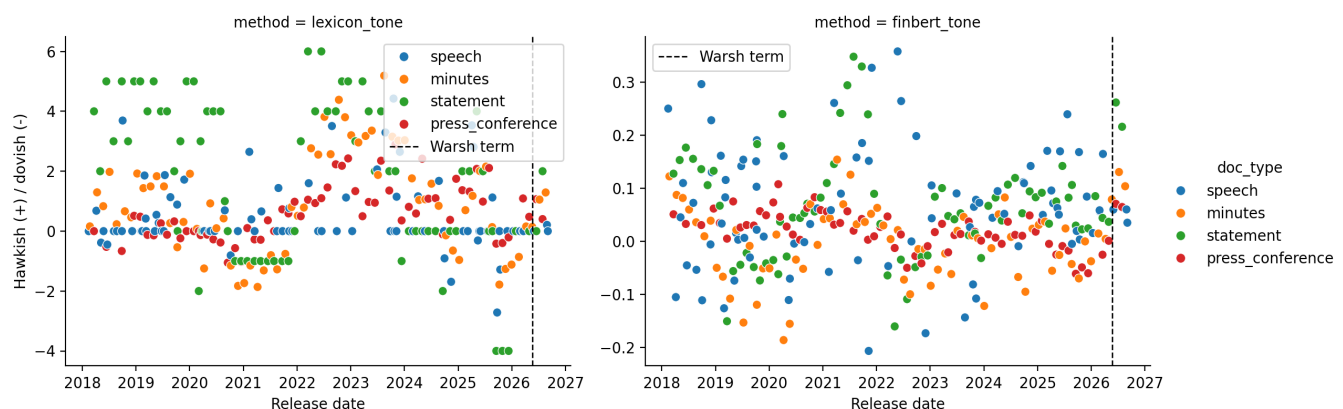
Tone is measured two ways: a monetary-policy hawkish/dovish word list normalized by document length, and sentence-level FinBERT sentiment (positive minus negative probability). The sample is expanded, but records with date-only or assumed timestamps must be manually checked before a final event study.

Table 1. Documents collected, by type and Chair

Document type	Powell	Warsh
Minutes	67	2
Press-conference transcripts	63	2
Chair speeches	86	3
Post-meeting statements	72	2

Table 1 now covers all three required communication classes. It still requires a timestamp audit: 154 records do not have a directly verified release time.

Figure 1. Hawkish/dovish tone over time by document type



The expanded figure contains both methods. The word-list series shows positive Warsh readings for the July statement (2.000), August minutes (1.291), and June press conference (1.065). FinBERT readings for the same documents are modestly positive (0.216, 0.104, and 0.070). The two methods broadly agree that recent communication is not strongly dovish, but their market-regression implications are not aligned.

Table 2. One-day market changes after each Warsh-era release

DXY and growth-minus-value are percentage changes. Treasury values and 10s2s are percentage-point changes. Speech timestamps shown at noon are provisional.

Release	Type	Lexicon	FinBERT	DXY	10s2s	1-year	Growth minus value
2026-05-31 12:00	Speech	0.000	0.060	0.293	-0.050	0.040	1.248
2026-06-17 14:00	Statement	0.000	0.261	0.553	-0.090	0.140	-0.258
2026-06-17 14:00	Press conference	1.065	0.070	0.553	-0.090	0.140	-0.258
2026-07-08 14:00	Minutes	1.053	0.130	-0.089	-0.010	0.000	1.506
2026-07-29 14:00	Press conference	0.401	0.065	-0.572	0.100	-0.050	-0.966
2026-07-29 14:00	Statement	2.000	0.216	-0.572	0.100	-0.050	-0.966
2026-08-19 14:00	Minutes	1.291	0.104	-0.823	-0.060	0.010	-0.842
2026-08-28 12:00	Speech	0.208	0.060	0.545	-0.080	0.110	-0.208
2026-09-03 12:00	Speech	0.000	0.035	-0.562	0.030	-0.050	1.025

The same-day statement and press-conference rows deliberately share a market move: they were released on the same announcement date. They should not be treated as independent events in a final inference exercise.

Table 3. One-day regressions with the 3-month bill control

Each row regresses the one-day outcome on one tone score and DGS3MO, using HC3 robust standard errors.

Indicator	Tone method	N	Tone beta	Tone p-value	DGS3MO beta	R-squared
DXY	Lexicon	297	-0.022	0.107	3.671	0.075
DXY	FinBERT	297	0.300	0.279	3.667	0.072
10s2s	Lexicon	297	-0.002	0.197	-0.401	0.092
10s2s	FinBERT	297	-0.017	0.508	-0.393	0.087
1-year Treasury yield	Lexicon	297	-0.001	0.479	0.876	0.290
1-year Treasury yield	FinBERT	297	0.006	0.821	0.878	0.289
Growth minus value	Lexicon	297	0.083	0.047	1.477	0.015
Growth minus value	FinBERT	297	-1.355	0.058	1.531	0.011

The word-list and FinBERT estimates for growth-minus-value have opposite signs, with both near conventional significance cutoffs. This conflicts with a strong single-method trading claim. All other text coefficients are statistically weak.

Comparison with the readings

Doh, Kim, and Yang (2021) argue that qualitative statement language can move financial conditions independently of the target-rate decision. This project uses DGS3MO as a control in the same spirit, but its broad daily window and mixed document types produce weaker yield results. The disagreement between the lexicon and FinBERT results also reinforces their concern that measuring policy language is not straightforward. [Doh, Kim, and Yang (2021)](<https://www.kansascityfed.org/documents/7577/erv106n1dohkimyang.pdf>)

Doh, Song, and Yang separate tone, novelty, and the surprise component using alternative FOMC statements and high-frequency data. This project improves on a single word list by adding FinBERT, but it still uses absolute tone and daily returns. It cannot make their counterfactual or surprise-based causal claims. Adding novelty, a narrow event window, and clustered treatment of same-day statement/press-conference pairs would improve the design. [Doh, Song, and Yang](<https://www.kansascityfed.org/documents/5642/rwp20-14dohsongyang.pdf>)

The *Parsing the Fed* comparison is now complete for word-list and FinBERT, but not factor similarity. The cross-method disagreement makes factor similarity a useful next robustness check rather than a cosmetic addition.

September FOMC forecast

Rate decision

Cut	Hold	Hike
20%	75%	5%

The hold remains the modal outcome. Communication-tone evidence does not itself identify a policy-rule change, and the small Warsh sample argues against a more aggressive rate call.

Statement tone

There is a 65% probability that the September statement is more hawkish than the prior statement. Recent Warsh communications are generally positive on both scores, though the modest FinBERT magnitudes argue against a high-confidence hawkish surprise.

Market reaction

The current lexicon-based forecast uses the latest statement tone scenario.

Indicator	Probability it rises	Expected one-day change
DXY	43.4%	-0.070%
10s2s spread	42.4%	-0.008 percentage points
1-year Treasury yield	48.2%	-0.002 percentage points
Growth minus value	54.0%	+0.126%

Because FinBERT disagrees on growth-minus-value, these are scenario estimates, not a consensus forecast. A final report should show an explicit FinBERT-based forecast beside this table.

Recommendation and falsification

Position: take a small long-IWF/short-IWN position (long growth-minus-value) ahead of the meeting.

Why: the lexicon regression is positive and significant at 5%, and its forecast assigns growth-minus-value the highest probability of rising. The position is intentionally small because FinBERT's near-significant coefficient has the opposite sign, the Warsh sample is short, and timestamp uncertainty weakens precise event alignment.

What proves it wrong: close the trade if statement-day growth-minus-value is -0.50% or lower, or if the 1-year Treasury yield rises by at least 10 basis points. Either outcome contradicts the lexicon scenario and indicates a more restrictive market interpretation.

Submission checklist

- Add factor-similarity tone scoring and report it alongside the word-list and FinBERT methods.
- Aggregate same-day communications into a single event for Table 3 regressions and the market forecast.
- Generate and embed Figure 1.
- Export the current report to PDF.
- Rerun Cells 1-14 after the code changes so Tables 1-3, Figure 1, and the PDF use the factor-similarity and event-level results.
- Review each remaining `date\_only\_manual\_time\_check` record and add a verified release time in `DOCUMENT\_OVERRIDES`; the code now uses official page metadata when it provides a timestamp.